

Compact Kähler 3-folds with the action of an abelian group of maximal dynamical rank

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Contents

1. Dynamical degree & Entropy
2. Background & Previous results
3. Projective setting
4. Main result and proofs
5. Open questions

Dynamical degree & Entropy

- (X, ω) : a compact Kähler manifold of dim n with a Kähler form ω .
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- The p -th dynamical degree $d_p(g)$ of $g \in \text{Aut}(X)$ is defined as

$$d_p(g) := \max\{|\lambda| : \lambda \text{ is an eigenvalue of } g^*|_{H^{p,p}(X, \mathbb{C})}\},$$

i.e., the spectral radius of the induced linear map $g^*|_{H^{p,p}(X, \mathbb{C})}$.

$$d_p(g) = \lim_{k \rightarrow \infty} \left(\int_X (g^k)^*(\omega^p) \wedge \omega^{n-p} \right)^{1/k}$$

- $h_a(g) := \max_{0 \leq p \leq n} \log d_p(g)$: the algebraic entropy of g .

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Theorem (Gromov and Yomdin)

The algebraic entropy coincides with the topological entropy $h_t(g)$, a dynamical invariant measuring the divergence of the orbits.

Dynamical degree & Entropy

Property – Concavity of dynamical degrees (Khovanskii and Teissier)

$$\exists u \leq v \text{ s.t. } 1 = d_0(g) < \cdots < d_u(g) = \cdots = d_v(g) > \cdots > d_n(g) = 1.$$

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- $g \in \text{Aut}(X)$ is of **positive entropy**, if $h_t(g) > 0$, or equivalently $d_1(g) > 1$. A subgroup $G \subseteq \text{Aut}(X)$ is of positive entropy, if every non-trivial element in G is of positive entropy (Dinh-Sibony, 2004).

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Definition (Dynamical degree in the singular setting)

Let X be a projective variety (or compact Kähler space).

- $d_1(g) :=$ the spectral radius of g^* on the Néron-Severi space $\text{NS}_{\mathbb{R}}(X)$ (or Bott-Chern cohomology space $H_{\text{BC}}^{1,1}(X)$).
- g is of **positive entropy** if $d_1(g) > 1$, otherwise it is of **zero entropy**.

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- g is of **positive entropy** if $d_1(g) > 1$, otherwise it is of **zero entropy**.
- By (Dinh-Sibony, 2005), the dynamical degrees can also be defined for dominant meromorphic maps and they are bi-meromorphic invariants.

Background & Previous results

Motivated by Smale's work, Dinh and Sibony studied the structure for an abelian subgroup of $\text{Aut}(X)$.

Theorem (Dinh-Sibony, 2004)

Let $G \subseteq \text{Aut}(X)$ be an abelian subgroup where X is a compact Kähler manifold of $\dim n \geq 2$. Then

- The subset $U \subseteq G$ of the elements of zero entropy is a subgroup.
- There is a free abelian subgroup $H \subseteq G$ which is of positive entropy and has rank $\leq n - 1$ such that $G \cong U \times H$.
- If $\text{rank } H = n - 1$ (being maximal), the group U is finite.

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Example: complex torus; (Dinh-Sibony, 2004)

The rank estimates in the above theorem are optimal. As mentioned in (Dinh-Sibony, 2004), for each n , there is an n -dim complex torus T such that $\text{Aut}(T)$ contains $G \cong \mathbb{Z}^{\oplus n-1}$, of positive entropy.

Background & Previous results

Zhang proved a theorem of Tits type for compact Kähler manifolds; meanwhile, Dinh-Sibony's rank estimates were thus extended.

Theorem (Zhang, 2009; cf. Campana-Wang-Zhang, 2014)

Let $G \subseteq \operatorname{Aut}(X)$ be a subgroup where X is a compact Kähler manifold of $\dim n \geq 2$. Then either G contains a free non-abelian subgroup $\mathbb{Z} * \mathbb{Z}$, or there is a finite-index solvable subgroup H such that the subset N of zero entropy elements of H is normal in H and H/N is free abelian of rank $\leq n - 1$. In particular, $\operatorname{Aut}(X)$ satisfies Tits alternative.

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- Tits' theorem: every linear group G over $\mathbb{R}$ satisfies Tits alternative, i.e., any subgroup of G either has a free non-abelian subgroup, or is virtually solvable (possesses a finite-index solvable subgroup).
- Tits alternative for $\operatorname{Aut}(X)$ was first conjectured in the joint work (Keum-Oguiso-Zhang, 2009).
- (Hu, 2020) extends Zhang's Tits alternative to $\operatorname{char} > 0$.

Background & Previous results

Main question (Dinh-Sibony, 2004; Dinh-Hu-Zhang, 2015)

Classify n -dim compact Kähler manifolds admitting a free abelian group of automorphisms of rank r ($r \leq n - 1$) which is of positive entropy.

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Theorem (surface case; Cantat, 1999)

If X is a smooth complex compact surface (Kähler or not) admitting an automorphism of positive entropy, then X is either a complex torus, an Enriques surface, a K3 surface or a non-minimal rational surface.

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Theorem (Projective case with maximal dynamical rank; Zhang, 2016)

Let X be a normal projective variety of $\dim n \geq 3$ admitting the action of the group $G := \mathbb{Z}^{\oplus n-1} \subseteq \text{Aut}(X)$ of positive entropy. If X is not rationally connected, then with G replaced by a finite-index subgroup, X is birational to a quotient of a complex torus (G -equivariantly).

Background & Previous results

- Zhang also treated the case when G is virtually solvable.
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Example: rational 3-fold; (Oguiso, 2014), (Zhang, 2016)

There exists a rational 3-fold X with at worst klt singularities such that (i) X is a quasi-étale quotient of a 3-dim abelian variety and (ii) X admits an abelian subgroup $G \cong \mathbb{Z}^{\oplus 2} \subseteq \text{Aut}(X)$, which is of positive entropy and of maximal dynamical rank.

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Definition (G -equivariancy)

A rational (resp. meromorphic) map $(X, G) \dashrightarrow Y$ of projective varieties (resp. compact Kähler spaces) is **G -equivariant** if G descends to a biregular action $G|_Y$ on Y .

Background & Previous results

The existence of the G -equivariant resolution has been confirmed.

Equivariant resolution (Hironaka, 1964; Włodarczyk, 2009)

Let X be an analytic space defined over $\mathbb{C}$. There exists a canonical desingularization of X that is a manifold $\tilde{X}$ together with a proper bimeromorphic holomorphic map $\text{res}_X : \tilde{X} \rightarrow X$ such that

- $\text{res}_X : \tilde{X} \rightarrow X$ is an isomorphism over the nonsingular locus of X .
- The inverse image of the singular locus $\text{res}_X^{-1}(X_{\text{Sing}})$ is a simple normal crossing divisor.
- res_X is functorial with respect to local analytic isomorphisms. For any local analytic isomorphism $\phi : X' \rightarrow X$ there is a natural lifting $\tilde{\phi} : \tilde{X}' \rightarrow \tilde{X}$ which is a local analytic isomorphism.

Remark: The existence of resolutions of singularities of analytic spaces (in $\text{char} = 0$) was confirmed by Hironaka. The proof has been simplified by Bierstone-Milman and Villamayor in their series of works.

Projective setting (Zhang's program)

The property of G -equivariant fibration (Zhang, 2009; Hu, 2020)

Let (X, G) be a pair where $G \subseteq \text{Aut}(X)$ is a free abelian group of positive entropy. If there is a G -equivariant surjective morphism $\pi : X \rightarrow Y$ of projective varieties (or compact Kähler spaces) with $\dim X > \dim Y > 0$, then $\text{dr}(G|_X) \leq \text{dr}(G|_Y) + \dim X - \dim Y - 1$.

($\text{dr}(G|_Y)$ = the rank of positive entropy part)

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A brief review of Zhang's program:

- If X is uniruled, $G|_X$ descends to biregular $G|_Y$ via Nakayama's special MRC fibration $X \dashrightarrow Y$. Applying the above property to a G -equivariant resolution, one can show X is rationally connected.

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Remark: Barlet's analytic cycle theory $\Rightarrow$ R.C. of uniruled Kähler 3-folds (with a free abelian group of maximal rank); cf. (Brunella, 2006).

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One needs an important nef and big class to ensure the G -equivariancy.

Theorem (Dinh-Sibony, 2004; Zhang, 2009)

Let X be a normal projective variety (or a compact Kähler manifold) of $\dim n$ and $G = \mathbb{Z}^{\oplus r} \subseteq \text{Aut}(X)$ a free abelian group of positive entropy ($r \leq n - 1$). Then there are nef $\mathbb{R}$ -Cartier divisors (resp. nef $(1, 1)$ -classes) $L_1, \dots, L_{r+1}$ such that

- each L_i is a common eigenvector of G , i.e., there are $r + 1$ characters $\chi_i : G \rightarrow \mathbb{R}_{>0}$ such that $g^* L_i = \chi_i(g) L_i$ for any $g \in G$; and
- $L_1 \cdots L_{r+1} \neq 0$ as an element in $H^{r+1, r+1}(X, \mathbb{R})$.

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- In particular, if $r = n - 1$ (being maximal), then we get a nef and big $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor (resp. $(1, 1)$ -class) $A := \sum_{i=1}^n L_i$.
 - (Property of A ; Zhang, 2016) For every G -periodic (k, k) -class $\alpha \in H^{k, k}(X, \mathbb{R})$, $A^{n-k} \cdot \alpha = 0$.

Projective setting

Using such A , we continue to recall Zhang's program for the pure case.

Proof of the G -equivariancy (Zhang, 2016)

- Finiteness of $(K_X + A)$ -neg. ext. rays $\Rightarrow$ Assume each $(K_X + A)$ -neg. ext. ray is A -trivial $\Rightarrow$ the G -image of a $(K_X + A)$ -neg. ext. ray is still $(K_X + A)$ -negative $\Rightarrow$ the G -equivariancy $\Rightarrow A$ is the pull-back of a nef $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor from downstairs by the cone theorem.

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- (BCHM, 2010) $\Rightarrow$ such G -MMP with scaling will terminate in finitely many steps $\Rightarrow$ the end product Y with $K_Y + A_Y$ being nef $\Rightarrow$ may assume $K_Y + A_Y$ is nef and big.

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- σ -decomposition for $K_Z \Rightarrow K_Z \sim_{\mathbb{Q}} 0$ and A_Z is thus ample.
- $A_Z^{n-2} \cdot c_2(Z) = 0 \Rightarrow c_2(Z) = 0$ (Miyaoka, 1985) $\Rightarrow Z$ is a Q -abelian variety (Yau, 78; Greb-Kebekus-Peternell, 16; Lu-Taji, 18).

Main result and proofs

Hypothesis: (X, D) is a $\mathbb{Q}$ -factorial compact Kähler klt 3-fold pair. There is an action of a free abelian group $G := \mathbb{Z}^2 \subseteq \text{Aut}(X)$, of positive entropy and of maximal rank. Every irreducible component of the support of D is G -periodic.

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Theorem (Z., 2022)

Let (X, D, G) be a pair satisfying Hypothesis. If X is not rationally connected, then with G replaced by a finite-index subgroup, we have

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Theorem (Z., 2022)

Let (X, D, G) be a pair satisfying Hypothesis. If X is not rationally connected, then with G replaced by a finite-index subgroup, we have

- 1 There is a G -equivariant bimeromorphic map $X \dashrightarrow Z$ such that Z is a quasi-étale quotient of a complex 3-torus.

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- 2 $Z \cong T/H$ where H is a finite group acting freely outside a finite set of a complex 3-torus T . Moreover, the quotient morphism $T \rightarrow Z$ is also G -equivariant.
- 3 There is no positive dim G -periodic proper subvariety of Z ; in particular, the bimeromorphic map $X \dashrightarrow Z$ in (1) is holomorphic.

Main result and proofs

Rationally connected compact Kähler manifolds being projective, we immediately have the following corollary.

Corollary

If X is a non-algebraic compact Kähler 3-fold admitting an action of a free abelian group $G := \mathbb{Z}^2 \subseteq \text{Aut}(X)$ of positive entropy, then X is bimeromorphic to a quotient of a complex 3-torus, with the quotient morphism being (G -equivariantly) étale outside finitely many points.

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Applying (Dinh-Sibony, 2004) to $\pi^* \text{Nef}(X)$, the pull-back of the nef cone along a G -equivariant resolution $\tilde{X} \rightarrow X$, we get three G -common nef eigenvectors ξ_1, ξ_2, ξ_3 with the sum $\xi := \xi_1 + \xi_2 + \xi_3$ a nef and big class.

Question

Are there only finitely many $(K_X + \xi)$ -negative (or non-positive) extremal rays in $\overline{\text{NA}}(X)$? Are they generated by rational curves?

A brief review on the MMP for Kähler varieties ($\dim \leq 3$)

- (Fujino, 2021) Fujino established the MMP for $\mathbb{Q}$ -factorial log surfaces in Fujiki's class.
- (Filip-Tosatti, 2018) proved the base point free theorem in $\dim 2$.
- (Höring-Peternell, 2015) & (Höring-Peternell, 2016) established the MMP for $\mathbb{Q}$ -factorial terminal Kähler 3-folds.
- (Campana-Höring-Peternell, 2016; 2021) proved the abundance for compact Kähler terminal 3-folds except that X is a simple 3-fold of Kodaira dim zero.
- (Das-Ou, 2022) proved the log abundance for compact Kähler log canonical 3-fold pair (X, Δ) with $K_X + \Delta$ being nef and the num dim $\nu(K_X + \Delta) \neq 2$.
- (Campana-Höring-Peternell, 2016), (Das-Ou, 2022) and (Das-Hacon, 2022) confirmed existence of the MMP for dlt pairs on Kähler 3-folds.
- (Das-Hacon, 2022) proved the base point free theorem on Kähler 3-folds; cf. (Höring, 2018) and (Tosatti-Zhang, 2018).

Main result and proofs

Proposition (finiteness of $(K_X + \xi)$ -negative extremal rays; Z., 2022)

Let X be a normal $\mathbb{Q}$ -factorial compact Kähler 3-fold and Δ an effective $\mathbb{Q}$ -divisor such that (X, Δ) has only klt singularities. Let α be a nef and big class on X . Suppose that one of the following conditions holds.

- ① $K_X + \Delta$ is pseudo-effective; or
- ② $K_X + \Delta$ is not pseudo-effective but $K_X + \Delta + \alpha$ is a big class.

Then there are only finitely many $(K_X + \Delta + \alpha)$ -negative extremal rays, all of which are generated by rational curves.

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- Our proof relies on the cone theorem. Under the above condition, there is a fixed number d , such that every $(K_X + \Delta)$ -negative extremal ray R is generated by a curve ℓ with $-(K_X + \Delta) \cdot \ell \leq d$.
- The cone theorem (✓) for the non-uniruled 3-fold case, but there are some differences with the projective case when X is uniruled.

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Proposition (End product of the G -equivariant $(K_X + \xi)$ -MMP)

If $K_X + \xi$ is nef, there is a G -equivariant bimeromorphic map $\tau : X \rightarrow Y$ such that Y is a quotient of a torus.

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- Base point free theorem (Das-Hacon, 2022) $\Rightarrow$ there is a holomorphic map $\tau : X \rightarrow Y$ with connected fibre such that $K_X + \xi = \tau^*\beta$ for some Kähler class β on Y .
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- τ is G -equivariant, but the proof is a bit more troublesome when compared with the projective case. We skip the technical proof here.
- τ is ξ -trivial $\Rightarrow \xi = \tau^*\xi_Y \Rightarrow K_X = \tau^*K_Y$ with $\beta = K_Y + \xi_Y$ Kähler.
- (Lemma) $\text{Null}(\xi_Y)$ (union of the positive dim proper analytic subvariety V such that $\xi_Y|_V$ is not big) = union of the G -periodic proper analytic subvariety.

Main result and proofs

- Let $\pi_1 : Y_1 \rightarrow Y$ be a G -equivariant resolution with the non-Kähler loci $E_{\text{nK}}(\pi_1^* \xi_Y) = \text{Null}(\pi_1^* \xi_Y)$ (Collins and Tosatti, 2015).

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- By (Demailly, 1992) and (Boucksom, 2004), we may choose a Kähler current T in $\pi_1^* \xi_Y$ with analytic singularities along $E_{\text{nK}}(\pi_1^* \xi_Y)$.
- Blowing-up the ideal sheaf and taking Hironaka's resolution, we have $\pi_2 : Y_2 \rightarrow Y_1$ such that $\pi^* \xi_Y = \omega + E$ (with $\pi := \pi_1 \circ \pi_2$) where ω is a Kähler class and $E \geq 0$ is an $\mathbb{R}$ -divisor, $\text{Supp}(\pi_2(E)) = E_{\text{nK}}(\pi_1^* \xi_Y)$.

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- Hodge index theorem $\Rightarrow \xi_Y|_V \equiv 0$ for every positive dim G -periodic proper subvariety V .
- $\pi^* \xi_Y|_E \equiv 0 \Rightarrow \pi^* K_Y \cdot \omega^2 = 0$ (Hodge-Riemann) $\Rightarrow K_Y \equiv 0$, noting that K_Y is psuedo-effective $\Rightarrow \xi_Y$ is a Kähler class.

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- $c_2(Y) \cdot \xi_Y = 0$; $c_2(Y)$ the orbifold second Chern class (Graf-Kirschner, 2020) $\Rightarrow Y$ is a Q -complex 3-torus by (Graf-Kirschner, 2020) (in the Kähler setting). The proposition is proved. (✓)

Main result and proofs

Some remarks:

- The G -equivariance of the quotient morphism $T \rightarrow Y$ shares almost the same proof as in the projective setting, so we skip it.
- Graf and Kirschner's characterization of Q -complex 3-tori has been recently extended to arbitrary dim, assuming the smoothness in codim 2; cf. (Claudon-Graf-Guenancia, 2021).
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Summarize the proofs

- (1). Start from the initial X and run $(K_X + \xi)$ -MMP G -equivariantly.
- (2). End up with some X_n such that $K_{X_n} + \xi_{X_n}$ is nef.
- (3). Such X_n admits a G -equivariant bimeromorphic holomorphic map to a quotient Y of a complex torus T .

$$X \dashrightarrow \cdots \dashrightarrow X_n \rightarrow Y \cong T/F.$$

Open question

In the view of Dinh-Sibony and Zhang's works, we would like to know the general case when G is assumed to be a free abelian group with lower rank.

Question 1

Let X be either a compact Kähler manifold or a projective variety of dim n . Suppose X admits the action of a free abelian group $G \subseteq \text{Aut}(X)$, of positive entropy and of rank $r \leq n - 1$. Can we classify such X ?

(Hu-Li, 2020) partially answered the above question when $\text{rank } G = n - 2$.

In the view of Oguiso's example on the rational klt 3-fold being a quotient of an abelian variety, we finally ask the following question.

Question 2; cf. (Zhang, 2016)

Is there a pair (X, G) with G being of maximal rank such that X is R.C. but X is not bimeromorphic to a quasi-étale quotient of a complex torus?

Thank you for your attention!